

Grapevine Culture (Part B): Weed, Pest and Disease Control

Lesson Aim

Specify the techniques used in the culture of grape vines (Weed, Pest and Disease Control).

Weed, Pest and Disease Control

Weeds, pests and diseases can significantly reduce the quantity and quality of yields in grape production. For this reason, effective weed, pest and disease management is essential to the operation of a good vineyard. Effective management relies on the correct diagnosis of problems and preventative controls.

Weed growth can have a significant impact on crop production. Weeds compete for nutrients, water and light, and can frequently be more vigorous than vines – particularly when vines are young. Weeds can also harbour pests and diseases. Some weed species are considered 'noxious' in some countries, and must, by law, be controlled.

Pests that damage grape vines range from small nematodes and insects to birds, rabbits, and larger mammals such as deer, cattle and wallabies. Problems caused by pests include eating the fruit, buds, leaves and fruit, trampling on vines, laying eggs in plant tissues, and spreading diseases. Diseases that infect and damage the fruit and vines are caused by bacteria, viruses or fungi.

Some pests and diseases are threats to the viticulture industry around the world. These pests and diseases have been the focus of much research on control mechanisms. Various mechanisms, such as grafting, gene technology and testing propagation material for viral infection are used in pest and disease control. Movement of grapevines and propagation stock from region to region is often restricted.

Crop damage can also result from nutrient deficiency or toxicity, excess water, inadequate water, and other environmental conditions. Identifying the cause of poor crop performance is an essential step in remedying the problem.

Weeds In Vineyards

Weeds can be defined as plants growing where they are not wanted. Weeds compete for light, nutrients, water and space. They can harbour pests and diseases, and make management tasks such as harvesting more difficult.

Ideally weed problems should be dealt with before vine establishment. Vines planted in clean, weed-free soil will quickly establish and produce vigorous growth. However even the cleanest vineyards require ongoing weed control as dormant weed seeds germinate and new weeds are brought into the vineyard.

There are many different ways of controlling weed growth, and the effectiveness of each technique is related to the varieties of weeds being controlled. Herbicides are widely used but other techniques such as cultivation, mulching, green manures, and fumigation can all be effective in different circumstances.

?

Early weed control is very important. As weeds grow larger, their effect becomes more significant. They also become much more difficult to control. Weeds – in particular annual weeds – should be prevented from setting seed. Removing weeds when they are still in an early stage of growth can be effective because this minimises competition and often juvenile plants are easier to kill.

The control method selected usually involves knowing what the weed is, and understanding its growth habit. Most adult weeds are easily recognisable but identifying young weeds can be difficult. At this early stage of growth, many species look much the same – they may be the same size, and have similar leaf shapes and colours. It is only as they mature and reach the adult stage that they develop distinctive characteristics such as flowers, which assist identification. With practice, growers soon learn to recognise even very small weeds, allowing them to select the most appropriate control method.

General Methods of Controlling Weed Problems

Restricting the Spread of Weeds

Weeds can be brought into the vineyard by wind, on dirty shoes, on vehicles, in soil, in plant containers, on tools, by birds and animals, and in manure or straw. Be mindful of these things and, wherever possible, try to avoid bringing weeds into your property.

Mowing

Regular mowing will prevent tall-growing annual weeds from flowering and setting seeds. It also gradually weakens perennial weeds as their food reserves become depleted.

Mulching

Mulches are used to smother existing weeds and to prevent weed seeds from germinating. A thick layer (10-12cm) of organic mulch will exclude many annual weeds and will have the added advantages of conserving soil moisture, protecting the soil from temperature extremes, and improving soil nutrition and structure as they break down.



Weeds controlled by mulch mat (material) together with chemical spraying.

Hand Weeding

This is most effective against broad-leaved annual weeds. It is time-consuming and, for many weeds which will regrow from a tiny piece of root, corm or tuber, it is not sufficiently effective to make any real long-term difference.

Chemicals

Herbicides are highly effective although over-use of agricultural chemicals has led to some concerns regarding their effect on the environment and on human health. Herbicides have different effects on weeds – choose the right chemical for the weed and its stage of growth.

Contact herbicide may kill only the parts of the plant to which they are applied (i.e. if they only touch the top half of a leaf, they may only kill that top half, and the bottom half of the leaf remains alive). Contact herbicides may be used to kill leafy above-ground growth on some annual weeds without affecting the adjacent vines (e.g. paraquat and diquat).

In some countries, it is now illegal to use many chemicals which were commonly used in the past. Do not assume the chemicals mentioned in this course are appropriate everywhere. Through our examples, you can gain a broad understanding of the mode of action of certain chemicals; but before choosing to use chemicals, you must ensure that a chemical is licensed (a) for use in your locality and (b) licensed for use on controlling the particular weed species in a vineyard situation.

Systemic herbicides will be absorbed into plant tissue and carried throughout the entire plant killing all parts of the plant. Typical systemic herbicides include glyphosate, 2,4- D and Mecaprop.

Pre-emergent herbicides kill weed seeds as they start to germinate.

These can be applied to an area in late winter before weed growth commences. Some common pre-emergent herbicides include bensulide, DCPA, and siduron.

Selective herbicides are those which kill one type of plant without affecting another type of plant. Certain chemicals can be used to kill some types of weeds in a planted area, without affecting others (e.g. 2,4-D; Dicamba, MCPA).

Fumigation

Chemical fumigation can be used to completely sterilise an area, killing weed seeds and ALL plants in an area. After fumigation, the site is generally rested for three days or more to get rid of all chemical residues. This method would perhaps only be used to clean up a major problem before planting a new vineyard block.

Burning

Flame guns can be used to burn weeds out in spot control. Burning is more applicable for control of weeds on fence lines and other places adjacent to vines which are sources of weed seeds.

Nutrition Management

Changing the balance of the soil nutrient status can make the soil more favourable for the preferred plants and less favourable for the weeds. For example, the weed Onion Grass (*Romulea rosea*) may be deterred from growing in soil fertilised with superphosphate; similarly, soil fertilised with nitrogen can deter legume weeds.

Safety Procedures When Using Agricultural Chemicals

Laws in Relation to Chemical Use

Anyone using agricultural chemicals must be aware of the regulations concerning the spraying, storage, transportation and disposal of chemicals. Ignorance is no longer an excuse! Each country, state, or region may enforce different government regulations that control chemical use. It is advised that you become acquainted with your local laws and relevant Acts. You should be aware of the regulations in your country as well as those for your state/region.

The status of individual chemicals is constantly reviewed worldwide; those listed here may have undergone or be undergoing a change in status in your region.

Material Safety Data Sheets (MSDS), Safety Data Sheets (SDS) or Pesticide Information Sheets are available for all manufactured chemicals and must be supplied with them. You should ensure that you have a copy of each sheet for every chemical that you have in use or storage.



Important Guidelines for Using Chemicals

1. ONLY use chemicals when they are actually needed.
2. Use the correct chemical for the job at hand; if unsure, seek advice.
3. Always read the label, and the product information sheets (if available).
4. Wear protective clothing at ALL times.
5. Use the correct pesticide application equipment.
6. Don't spray on windy or very hot days.
7. Warn other people in the area that you are going to spray (and have sprayed).
8. Wash out all spray equipment thoroughly when finished.
9. Do not eat or smoke while spraying.
10. Wash all protective clothing thoroughly after spraying.
11. Wash yourself thoroughly after spraying especially the hands.
12. Store the spray equipment and chemicals in a safe, locked place.
13. Dispose of empty pesticide containers according to the label instructions.
14. Record all details of your spraying.

As stated above, some chemicals listed may be restricted or banned in some countries. Chemicals are constantly under review and various countries or states have differing laws governing their use. It is imperative that you research chemicals before using them. Remember that a product may have restricted use i.e. it may only be used to control a particular weed or pest in a particular crop/situation and it may not be used to control the same weed or pest in a different crop/situation, e.g. in the UK a product with the active ingredient carbetamide may be used to control broad-leaved weeds in Brassicas grown for leaf and flower-head production, but not on salad Brassicas.

Keeping Records of Chemical Use

Records of chemical use are essential. The record will help you to:

- Improve pest control practices and avoid unnecessary pesticide use.
- Compare applications made with results achieved.
- Purchase only the amounts of pesticides needed.
- Reduce inventory carry over
- Establish the cause of errors, if they occur
- Establish proof that you used recommended procedures if indemnity payments are involved.

What Information Should be Kept?

- Varieties of grapes treated.
- Pests (Insects/Fungus/Weeds) treated.
- Location and size of area being treated.
- Time of day, date and year.
- Type of equipment used.
- Pesticide used including name, type of formulation, trade name, manufacturer and batch number.
- Amount used per hectare or per 100 litres of water.
- Amount of active constituent (i.e. chemical) per hectare or per 100 litres of water.
- Stage of vine development (size of plant/year of growth/pre-fruiting/post-harvest etc).
- Pest/weed situation (e.g. severe, mild).
- Weather conditions, temperature, wind, rainfall
- Whether the chemical was watered in afterward
- Results of application: how long before pest was controlled; how well it worked.

Types of Herbicides

Most herbicides fall into one of the groups listed below. Although all groups may not be relevant to grape vines and their production, it is useful to understand this system of classification because their characteristics are likely to be shared with different chemicals in the same group. If you know one chemical in a group is either very safe, or very poisonous, it is likely that other chemicals you encounter within the same group will have a similar hazard level. If you know that a chemical within a particular group is a systemic action, then it is likely that other chemicals in that group will also be systemic, at least to some degree.

Herbicide names have been assigned by the International Organisation for Standardisation (ISO), in accordance with an established system of nomenclature. Names used here are the ISO-approved standard names of herbicides: names used for herbicides that do not have ISO names are those from national and international bodies.

Group 1 ALIPHATICS

The definition of Aliphatic is scientific and indicates that there are no molecular rings in the structural formula of the chemical. Aliphatics are organic chemicals, and include such herbicides as Dalapon, Methyl bromide, TCA (halogenated aliphatic herbicides):

Group 2 AMIDES and ANILIDES

Most amides are selective herbicides that are used as pre-emergent or pre-plant applications. There are three groups:

- Soil-applied Chloroacetamides and Chloroacetanilides. Usually only result in pre-emergent kill on germinating seeds: some will kill certain very young plants, e.g. CDAA, allidochlor, Isoxaben, Propyzamide(amides), Propachlor, Alachlor, Butachlor, Metazachlor (chloroacetanilides).
- Other soil-applied Amides. These have little if any post-emergence effect. They generally control a broader variety of weed species than group 2a/ (above), e.g. Diphenamid, Fomesafen, Naptalam, Pronamide, Napropamide).
- Foliar-applied Amides and Anilides, e.g. Propanil which causes chlorosis and death on leaves of susceptible plant species, particularly certain annual grasses: Diflufenican: Mefluidide, Pentanochlor (anilides): Flufenacet (amide).

Group 3 BENZOICS

These chemicals are derived from benzoic acid, all containing at least two chlorine atoms. Virtually all are effective applied to foliage or soil (except chloramben - this has only a pre-emergence action). Examples: 2,3,6 -TBA, Dicamba, Chloramben.

Group 4 BIPYRIDILIUMS

These are ammonium relatives, (quaternary ammonium herbicides), heterocyclic organic compounds, with contact action on foliage, causing wilting and rapid deterioration of foliage. Examples: Paraquat, Diquat.

Group 5 CARBAMATES and CARBANILIDES

Herbicides derived from Carbamic Acid. There are two types within this group:

- Soil Applied Carbanilates, e.g. Protham, Chlorprotham
- Foliar Applied Carbamates and carbanilides, e.g. Asulam (carbamate): Barban, Desmedipham, Phenmedipham, (carbanilates).

Group 6 DINITROANILINES

Most are yellow, orange crystalline solids which are not very soluble in water. They are normally incorporated into the soil as granular herbicides. Their effect is a selective one. Examples: Butralin, Fluchloralin, Oryzalin, Pendimethalin, Profluralin, Trifluralin.

Group 7 DIPHENYL ETHERS



Used mainly to control annual weeds in certain fruit, vegetable and field crops. Examples: Acifluorfen, Bifenox, Fomesafen, Nitrofen, Oxyfluorfen (nitrophenyl ethers).

Group 8 NITRILES

These are organic compounds containing a carbon-nitrogen group. They include both soil and foliar applied herbicides. Examples: Dichlobenil, Bromoxynil, Ioxynil.

Group 9 PHENOXY

Chemicals, which have a phenyl ring attached to an oxygen molecule, which in turn is attached to an aliphatic acid. Examples: 2,4-D; 2,4-DB; MCPA; 2,4,5-T; Diclofop-methyl (aryloxyphenoxypropionic herbicide).

Group 10 THIOCARBAMATES

Derived from carbamic acid, but with one oxygen atom replaced by a sulphur atom. Most are relatively volatile and need to be mixed with the soil by cultivation immediately after application, otherwise they disappear. Examples: Butylate, Cycloate, Di-allate, EPTC, Molinate, Pebulate, Thiobencarb, Tri-allate, Vernolate.

Group 11 TRIAZINES

Heterocyclic nitrogen derivatives; mainly used as selective herbicides on crops. Also used in total weed control. Examples: Ametryn, Prometryn, Terbutryn (methylthiotriazines); Atrazine, Cyanazine, Cyprazine, Propazine, Simazine (chlorotriazines); Prometon (methoxytriazine).

Group 12 UREAS and URACILS

These are substituted ureas and urea-type herbicides. Their use and action on the plant tends to be similar with all types of ureas and uracils. They are usually non-selective at high rates. At low rates some may be selective. Examples: Bromacil, Terbacil (uracils), Diuron, Fluometuron, Linuron, Monuron, Siduron, (phenylureas): Tebuthiuron (thiadiazolyurea).

Group 13 OTHER ORGANIC HERBICIDES

This includes a range of different organic herbicides, some of which do not fall into any specific group.

Examples:

Amitrole

- Triazole herbicide
- Oral LD50 is 24,600.
- Used as foliar spray.
- Controls annual weeds and certain perennials.
- Often used with simazine as a total control on paths, fence lines etc.

Bensulide

- Organophosphorus herbicide
- Oral LD50 is 770.
- Persists in soils for up to 12 months.
- Does not work on soils containing high quantities of organic matter.
- Controls annual grasses and selected broadleaved weeds in established lawns.
- Also used for selective weed control in some ornamental gardens and certain crops.

Petroleum Oil



- These are hydrocarbons, derived from petroleum.
- A commonly used example is kerosene.
- These were used extensively in the past as contact herbicides for either selective or total control, however use in recent decades has diminished greatly.

Other herbicides in this group include Endothal/Endothal, Fenac (fenac in Canada and USA, chlorfenac elsewhere), Glyphosate (organophosphorous herbicide): Acrolein, Fluridone, Methazole, Prosulfalin (unclassified herbicides): Dinoseb (dinitrophenol herbicide): Norflurazon, Pyrazon (pyridazinone herbicides): Perfluidone (sulfonanilide herbicide): Picloram (picolinic acid/pyridine herbicide): DSMA, MAA, MAMA, MSMA (arsenical herbicides).

Group 14 INORGANIC HERBICIDES

These are chemicals which contain no carbon atoms. The main herbicides in this group are the borates and chlorates. Others are the cyanates and ammonium sulfamate (formerly sulphamate). Borates are herbicides which are boron salts. They have LD50's from 2000 to 5560. They are generally non-selective, used for total weed control and are long lasting. These days, inorganic herbicides are generally not recommended because of their persistence in the soil.

Weed Management in Vineyards

Weed control is essential in new vineyards, where it enhances establishment of the vines, and in mature vineyards, where it will improve growth and yield of established vines.

No one weed management program can be prescribed for vineyards. Weed control practices are varied and will differ from vineyard to vineyard and country to country. Usually several techniques will be employed in the same vineyard, depending on such things as the weed species, age of vines, soil conditions, irrigation practices and topography. Some growers are electing to use 'organic' (non-chemical) methods; others use a combination of herbicides and non-chemical weed controls.

Any weed management program should start pre-planting. It is far easier to control the more noxious and persistent weeds (perennials) before the vines are planted. The first four years of a vine's growth are most critical in respect of competition from weeds – competition for water, nutrients and sunlight. In addition to direct competition, weed growth provides habitat for animals, pests and diseases all of which may damage the vines. In dry periods, dry weed growth is a fire hazard.

Even after vine establishment, weed growth will interfere with cultural practices and harvest, eg. weed foliage can disrupt the pattern of water application from low-volume irrigation spray emitters; frequent cultivation to remove weeds at the base of vines can lead to root injury, which in turn allows infestation by pest and disease.

Soil type greatly influences weed growth. Generally speaking, annual weed species such as Crabgrass (*Panicum* spp.) and puncture vine (*Tribulis terrestris*) and the perennials Bermuda buttercup (*Oxalis pes-caprae*), Bermuda grass (*Cynodon dactylon*) and Johnson grass (*Sorghum halapense*) are more troublesome on light-textured soils. Perennials such as curly dock (*Rumex crispus*) and field bindweed (*Convolvulus arvensis*) are common on heavier-textured soils.

The soil type also influences the herbicide used. For example, less pre-emergent herbicide is required for effective control where soils are light; and lower levels of herbicide are generally required on sandy soils or those with low organic matter content.

When using herbicides in vineyards, combinations or sequential applications of different herbicides will be required for effective and economical control. Herbicides are usually split into two general groups: pre-emergent (on bare soil against germinating seedlings) and post-emergent (against weeds already growing). Post-emergent herbicide pose more risk to growing vines as spray-drift/contamination will likely damage the vine as well as the weed.

Weed Management Before Planting

It is important to control both annual and perennial weeds before planting vines, to avoid competition. Established stands of perennial weeds will be clearly evident but annual weeds can pose equally significant competition.

Non-Chemical Controls

Cultivating and irrigating the soil, thus encouraging annual weeds to germinate before planting the vines, is very effective. A minimum of two cycles of cultivation and irrigation is needed for effective control. Perennial weeds cannot be controlled this way.

In very dry conditions, if the soil is thoroughly cultivated, the rhizomes of perennials such as Bermuda grass (*Cynodon dactylon*) and Johnson grass (*Sorghum halapense*) will be chopped into small pieces which will quickly dry and die. If rain falls before the drying is complete, the rhizomes will germinate. This is not necessarily a bad thing – smaller weed plants from the smaller rhizomes can be more easily controlled with herbicide. Field bindweed (*Convolvulus arvensis*) can be reduced for 1-2 years following deep ploughing (45cm or 18in deep).

Soil solarisation is now a common non-chemical method of controlling weeds and soil-borne pests and diseases. Clear plastic sheeting is laid on moist soil when the ambient temperature is high. The sun's radiant energy is trapped in the soil by the plastic sheeting; during hot summer months the upper levels of the soil will be heated to temperatures that will kill many weed seedlings as well as pathogens (disease-causing organisms) and nematodes. Compared to cultivation, soil solarisation has the additional advantage of preserving the soil structure.

Chemical Controls

Pre-emergent or post-emergent herbicides may be used before planting. The residual period of a pre-emergent herbicide should not be so long as to preclude planting of the vines. In many cases, post-emergents are the safer option before planting (shorter residual periods), followed by use of pre-emergents after planting.

Weed-Management in New Vineyards

Weed-infested vineyards take longer after planting to become economically productive because of competition from the weeds for water, nutrients and sunlight. Any weed control must be carried out very carefully, particularly while the vines are establishing to prevent damage from chemicals and machinery.

Cultivation

If the grower is aiming to manage without herbicides, hoeing, cultivating or the use of weed knives (which penetrate the soil less than 5cm (2in) deep) are carried out several times during spring and summer. The areas between the rows must also be cultivated or mown. This operation will not be successful if the weeds are allowed to get large. To avoid injury to the vine, it is usually necessary to use hand-tools close to the stem. Mechanical equipment can be used between the rows.

Cover Crops

The choice of cover crop used will depend on your locality. The cover must not compete with the vines, but must be dense enough to smother the weeds. The cover can be left in situ for longer, by controlling its height by mowing, if that is appropriate for the crop chosen.

Mulching

Weeds in the vine row can be controlled with mulch. The mulch can be organic or synthetic. Mulch should be applied to clean (i.e. weed-free), moist soil. Mulch prevents annual weed seedling growth by blocking out light, and it provides a uniform soil moisture status which promotes better vine growth. The disadvantages of mulch can be that it provides a home for snakes (which may be dangerous to humans) or small mammals (which may damage vines) and that there may be new seed in organic mulch, perpetuating the weed problem. Mulches will not control perennial weed growth adequately.

Herbicides

Control carried out after planting but pre-bearing (fruiting) can be carried out using a pre-emergent herbicide as a 50cm (2ft) wide band along the vine row or as a 1m (3ft) diameter square or circle around each plant. Selective post-emergent herbicides can be used for both perennial and annual weeds. If a non-selective herbicide, such as glyphosate, is used extreme care must be taken to

prevent contamination of any green part of the vine (stems/leaves/shoots). Preferably, non-selective herbicides should only be used once the vine is sufficiently mature to have brown bark; in this case care must still be taken to prevent contamination of green shoots or leaves.

Weeds between the rows can be controlled by mowing as required to maintain a height lower than 15cm (6in) or cultivating as seeds germinate following rain or irrigation.

Weed Management in Established Vineyards

Once the vineyard is established (about four years) weed control is easier because the vines are usually more tolerant of herbicides. Cultivation or mowing is still used to control weeds between the rows. In the vine row, herbicide is applied as a basal treatment around the vine or in a band along the row.

Cultivation

Cultivation in the row is not recommended. The vine roots will have spread extensively and will be damaged by mechanical methods.

Flame Guns

Flaming can be used only around mature vines on young weeds. Fire is a potential risk.

Mulch

Mulches can be used in the same way as described in the previous section (Weed Management in New Vineyards). Organic mulc biodegrades quite quickly and becomes a perfect seed bed for weed seeds. Mulch must therefore be replenished regularly (at least annually).

Herbicides

Combinations of pre-emergent and post-emergent herbicides can be used after harvest, or as two applications, one in spring and one post-harvest. Where mulches are used, spot weed treatment may be possible, thus reducing the number of, and need for, herbicide applications. By monitoring which weed species are present, herbicide applications can be targeted and specific.

Cover Crops

In some countries it is possible to use late-seeding cereals, such as wheat or oats, as a cover crop. However, it is important to change the cover crop species annually to avoid the potential for pathogen build-up.

Integrated Pest Management (IPM) in Vineyards

Managing pests and diseases is a major aspect of viticulture operations. Until recently, chemical sprays provided the most reliable short-term control for a range of pests and diseases. However, repeated and intensive use of chemical sprays has resulted in a range of problems, including increased operational costs, risks to human health, long-term damage to the environment, and build-up of resistant pest and disease populations.

Many growers are now realising the long-term, sustainable benefits that can be gained by managing pests and diseases using a combination of compatible chemical, biological and cultural controls. This practise is known as Integrated Pest Management or IPM. The principle of IPM relies upon creating, as far as possible, an environment where there is a balance between sustainable environmental practices and profitable farming.

The concept is that:

- Nothing should be used to the detriment of the environment or to the extent that pests get accustomed to the method (as they may build up resistance);
- Each technique weakens the pest or disease. The overall effect is cumulative, and may be quite effective;
- Expensive controls (e.g. some costly chemicals) are used in limited quantities, reducing operational costs.

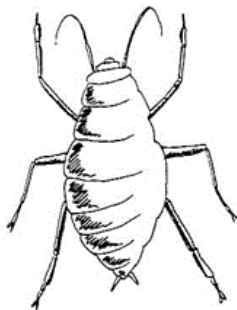
The basis for natural control is to regulate the environment to give nature the very best chance of keeping pest and disease problems in check. This may take some time to achieve and may result initially in higher losses until a balance is achieved. This is also very difficult to achieve for the grower surrounded by other properties where other means of pests and disease control are used.

Integrated pest management does allow the use of pesticides and herbicides, but only as part of an overall management program.

Pest Control in Vineyards

Pests of grapes include nematodes, insects such as the grape blossom midge, grape berry moth, leaf hopper and phylloxera, as well as birds and larger animals such as deer, sheep, cattle and other herbivores. Pests in a vineyard will differ, both in type and severity, depending on the locality. Some, such as the Grape Phylloxera (an aphid-like species, *Daktulosphaira vitifoliae*), are a significant global threat. Others may be a problem in one region but not another. Some of the more significant pests of grape vines are discussed below.

Remember: In some countries, it is now illegal to use many chemicals which were commonly used in the past. Do not assume the chemicals mentioned in this course are appropriate everywhere. Through our examples, you can gain a broad understanding of the mode of action of certain chemicals; but before choosing to use chemicals, you must ensure that a chemical is licensed (a) for use in your locality and (b) licensed for use on controlling the particular pest species in a vineyard situation.



Aphid

Grape Phylloxera

Grape phylloxera is native to the eastern United States of America, but has now been distributed by humans throughout the world. Phylloxera is a significant threat to vineyards worldwide, although phylloxera-free regions certainly exist. The United Kingdom represents one such area where phylloxera is not considered to be a major problem. The grape phylloxera are microscopic aphids, *Daktulosphaira vitifoliae*, that cause leaf galls, which are unsightly but cause little damage. Phylloxera also cause bulbous, yellow gall swellings on vine roots, which lead to decline of the vine through defoliation and reduced shoot growth. Hosts are both wild and cultivated vines.

Presence of phylloxera is easily identified by the leaf and root galls, which are usually about 6mm ($\frac{1}{4}$ in) in diameter and wart-like.

Phylloxera often is a complex pest as generations at different stages of their life cycle can exist simultaneously. There is a wingless form, which is very small and aphid-like, yellow-brown and pear shaped. The less visible winged form, is also aphid-like, but differs in that wings are held flat over the back. Neither form has the typical aphid cornicle (pipe-like structures on the top of the

abdomen).

The female hatches from an egg that has over-wintered on the grapevine wood. She migrates to a leaf where she produces a gall. In 15 days, at maturity, the female fills the gall with eggs and dies. Nymphs from these eggs, migrate to new leaves where they repeat the production of a gall and eggs. This may be repeated by up to seven generations in one summer. In the autumn the nymphs migrate to the grapevine roots, where they hibernate. In the following spring, the root galls are created and the cycle may be repeated year after year by these nymphs. Some of the root-inhabiting females lay eggs which develop into winged females. These migrate to the leaves and lay small male eggs and larger female eggs. After mating, the female lays a single fertilised egg which is the egg that will over-winter on the grapevine wood. The cycle then starts all over again. Leaf-inhabiting forms will not occur without the root-inhabiting form occurring also.

The nymphs can move from plant to plant along roots and by moving short distances along the soil surface. Individuals may occasionally travel onto aerial plant parts and be transported by wind or harvest machinery; they can also be spread on propagation material.

Management

The only completely effective means of control in severely infested areas is the use of resistant rootstocks. Phylloxera prefers heavy clay soils and chemical pesticides do not easily penetrate these. It is also difficult to eradicate without years of insecticide application because, although many are killed, the population seems able to regenerate rapidly. Prevention is therefore a better strategy, than cure.

Movement of propagation materials is restricted in many countries. Where it is permitted to move materials, these should be sterilised before being transported out of an infected region.

Whilst cuttings can be grown in uninfected areas, resistant rootstocks are required in infected areas. Evidence of phylloxera infestation should always be reported.

Monitoring and Treatment

It is often unclear whether or not phylloxera is present before it is too late. If weakened vines appear, these must be closely monitored, particularly if vines in one area of the vineyard show persistent weakening. It is wise to uncover the roots of vines around the perimeter of weakened areas; check the whitish-yellow, hooked feeder roots for galls. A hand lens should be sufficient to see the phylloxera in the gall.

Biological Controls

Environmental and root conditions have more bearing on phylloxera than natural predators. Levels of magnesium and phosphorus should be maintained in the soil: vines deficient in these nutrients are more susceptible to phylloxera. The use of predatory nematodes is being researched.

Cultural Controls

Use only clean propagating material and resistant rootstocks. Resistant rootstocks are frequently accepted as an organic management method, but you would need to confirm this with your local regulatory body.

Chemical Controls

Chemicals, when used, should be used as part of an integrated management system. Your local regulatory department will advise which chemicals are licensed for phylloxera control. There will be restrictions on the use of chemicals such as time intervals pre- and post- harvest, and time intervals between application and access to the vineyard. In France, during the 1860's, Bordeaux mixture (copper sulphate/lime mixture) was developed to control phylloxera. This product is still used in some countries to reduce (not eradicate) the effect of phylloxera.



Grape Berry Moth

The Grape Berry Moth (*Endopiza viteana*) is a small moth that, in its larval stage, attacks almost-ripe grapes and often binds them with silk. Damage caused by Grape Berry Moth can often lead to disease infection such as botrytis. The adult moth is about 6mm ($\frac{1}{4}$ in) long. It has a brown band across its middle, when at rest with folded wings, and the body is grey-blue. The rear part has brown markings. Pupae are found under a small flap, cut into the vine leaf; they are about 5mm ($\frac{1}{5}$ in) long and dark greenish-brown in colour.

Adults emerge just before blossom time and lay single eggs on fruit stems, which they cover with a fine web. Eggs will hatch in about 5 days. The larvae feed on blossoms and young fruit for about 21 days. They then move to the leaves, where they make a semi-circular slit, the flap of which they fold over themselves and pupate. After another 10-15 days the adult moths emerge. In only another 4-5 days the moths will again start laying eggs, so 2-3 generations may occur in one season. The larvae from the second and third generation use fine webbing to protect them as they pass from one berry to another, feeding within the berries. Pupae from these generations over-winter on the ground.

Management

Control of the Grape Berry Moth has traditionally been with chemicals. Integrated Pest Management (IPM) techniques such as pheromone traps to monitor and control populations, or disrupt breeding, can be effective.

Monitoring and Treatment

The obvious signs of Grape Berry Moth is webbing over the fruits and blossoms in conjunction with the distinctive leaf flap cocoons.

Cultural Controls

Leaf litter should be cleared to eliminate the over-wintering stage.

Chemical Controls

Chemical controls are problematic because the larvae appear before the grape blossoms have been pollinated. Insecticides will eliminate beneficial pollinators as well as the pest. Insecticides can be applied when the grapes are about the size of a small pea. It is almost impossible to achieve chemical control of the second generation, because sprays will not penetrate the clusters of enlarging grapes.

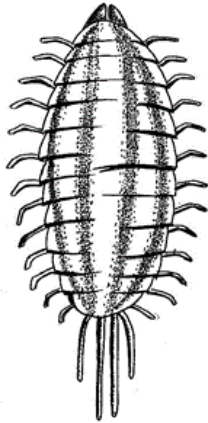
IPM

Pheromone traps which interrupt the mating patterns of the moths, used in conjunction with insecticide is the favoured control for Grape Berry Moth.

Grape Mealy Bug/Vine Mealy Bug

The Grape Mealy Bug (*Planococcus ficus*) is soft, oval, flat, distinctively segmented and covered with a white mealy wax extending into the spines. The pinkish body is visible through the wax. It is slightly smaller than the more common *Pseudococcus* Mealy Bugs, which will also infest grapevines. They may live on roots, under bark and move about on a plant according to seasonal conditions and the phase of growth the mealy bug is at. Most life stages of the mealy bug can be present on one vine, year round, depending on the growing locality. In lower winter temperatures, the nymphs only, are found at graft unions, on trunk pruning wounds and sometimes below spurs. As temperatures increase, populations increase and the mealy bugs move to the cordons and the canopy. The vine mealy bug has a support mechanism in the role of ants – the ants transport the mealy bugs from roots to aerial parts of the vine and continue to tend the mealy bugs throughout the growing season.

The Grape Mealy Bug causes similar damage to that of other mealy bugs. It produces honeydew which drips onto the grapes, and forms a substrate for black sooty mould. The quantity of honeydew may be large enough to resemble candle wax! The mealy bugs themselves crawl into bunches of grapes, making them unfit for consumption. Grape mealy bugs, like other mealy bugs, can transmit grape virus diseases.



Mealy Bug

Management

Grape mealy bug is largely spread by transportation of infected propagation materials. Whilst small infestations are relatively easy to treat, advanced infestations can be very difficult to eradicate. Infested vineyards can require annual pesticide treatments to keep problems under control.

Monitoring and Treatment

Pheromone traps are useful for monitoring the presence of grape mealy bugs in the vineyard. It is important to be sure that the mealy bugs in the trap are grape mealy bugs. The amount of honeydew produced by vine mealy bugs is much greater than other mealy bugs, providing ants are not present. Sooty mould will develop quickly in so much honeydew. Less severe infestations require active inspection under bark and around graft unions. The presence of ants may indicate the presence of vine mealy bugs.

Biological Controls

In California some success has been achieved using a species of parasitic wasp, *Anagyrus pseudococci*. The success has largely been due to the fact that these parasites are active late in the season and can parasitize the mealy bugs before they move to the lower parts of the vine out of reach of chemical sprays. Ants prevent the parasitism of this natural enemy, and must be controlled.

Cultural Controls

Because the female mealy bug is unable to fly, it must be transported by another means – this is usually human, equipment, birds or on vines at planting time. Propagating materials, machinery, winery waste, vines etc. must be clean.

Chemical Controls

Ideally this pest should be controlled early in the growing season when they are migrating. Pre-emptive sprays can be most effective, since by the time mealy bugs are detected, it is too late in the season to control them.

If the pest is found pre-harvest, but late in the season, chemicals may be applied before harvest, to start the process of complete eradication of the pest; Dimethoate is licensed for this use in some parts of the world. During harvest, precautions must be taken to prevent movement of the pest and immediately post-harvest a foliar insecticide spray is used, to catch the nymph mealy bugs.

before they move to the lower plant parts. The following season sequential treatments of different chemicals at different growth stages of the vine complete the eradication.

Alternatively, a yearly spray program can be used. This employs different chemical insecticides throughout the growing season.

All the chemical applications will be restricted by harvest and access intervals.

Grape Leafroller

Grape Leaf folder (*Desmia funeralis*) is a moth larva which although almost translucent appears green because ingested leaf tissue is visible through the body wall. The adult moth is almost black and has two white spots on the forewings and two white stripes on the abdomen. Usually the leafroller has 3 generations each year. Moths emerge in spring, lay single eggs on either surface of the leaves, although largest numbers are on the veins of the underside. After hatching, a group of larvae web two leaves together and feed voraciously. The larvae then roll the leaf edge and feed from the inside of the leaf edge. When ready to pupate, mature larvae construct an envelope like structure on the leaf edge.

The damage caused by the leaf folder is a significant reduction in leaf area. The first generation damage can usually be tolerated but damage by subsequent generations can lead to complete defoliation, but will, in any case, result in sunburnt, soft fruit. The larvae will also feed directly on the grapes.

Management

Parasites are significant in the control of leaf folder larvae.

Monitoring and Treatment

The first inspections should be frequent, to detect groups of larvae feeding between leaves. Treat as soon as leaf rolling starts.

Biological Controls

Several parasites will attack leaf folders. The most common, a wasp parasitoid *Bracon cushmani*, parasitizes the larvae and can frequently reduce the second and third generations of leafroller to below economic levels. There are at least two other hymenopteran (wasp) parasites and generalist predators such as lacewings will also attack leaf folders larvae.

Sprays of *Bacillus thuringiensis*, a parasitic bacterium, are also effective.

Nematodes

Nematodes are small, normally microscopic worms which attack and can burrow into plant tissue, commonly roots or leaves. They can cause distortion in growth (such as swellings), yellowing or dead patches.

Biological Control

French Marigolds (*Tagetes* spp.) can deter nematodes if planted within the vineyard rows (a chemical given off by the marigold roots deters soil nematodes).

Chemical Controls

Various chemicals are also used in nematode control, and soil ameliorants such as compost and lime can also reduce or prevent damage.

Leaf Hoppers

Leaf Hoppers are small sap-sucking insects that jump when disturbed. There are several species, including the Grape Leaf Hopper (*Erythroneura variabilis*) and others. Some are endemic to particular geographical locations; others are migratory. Leaf hoppers have few natural predators, although trials of parasitic wasps are proving successful in California. Cold and moist conditions in spring and autumn are damaging to leaf hopper populations. Control is usually achieved with contact pesticide. Care must be taken to ensure all leaf surfaces are covered. New research in Australia suggests that the debilitating vine disease, Australian grapevine yellows, is transmitted by Grape Leaf Hoppers.

Grapevine Rust Mite

Grapevine rust mite (*Calepitrimerus vitis*) is found in most grape growing areas. This mite will only become a problem if chemicals used to control other problems have reduced populations of predatory mites and insects. It causes damage to leaves and bunch stems. Mites are spread by several vectors including wind, insects and birds or on vine trimmings moved about the vineyard, and inside buds on vine cuttings used for propagation.

Adult Grapevine rust mites are tiny - around 0.2mm long. The head end is thicker than the tail end and has two pairs of legs. They are brownish in colour and wedge-shaped paler. During most of the growing season, adults, larvae and eggs can be seen on the main veins of upper and lower leaf surfaces. Sometimes a few mites will be found in buds, but if there are large numbers in the buds, these are more likely to be bud mites. Adults over winter under the vine bark. In spring, adult mites lay their eggs near the base of shoots. They progress along the shoots as they grow, feeding on the new leaves.

The first signs of Grapevine rust mite are evidenced by crinkling leaves, with tiny yellowish spots. The growing shoots may appear shorter. If the infestation continues, the crinkling of leaves at the growing tips is the most obvious sign. Because the early damage resembles other problems the mites are often overlooked. In the autumn, the leaves will exhibit the characteristic bronzing effect of mite infestation, as the earlier yellow spots become necrotic.

Biological Controls

Naturally-occurring beneficial insects and other predatory mite species often keep these pest mites under control with no need for chemical intervention, however if controls are required, for chemicals to be effective, application must be timed to target mites when they are exposed on the vines.

Chemical Controls

Grapevine rust mites can be controlled with sprays such as sulphur. Application prior to bud-burst can achieve good control. Applications of chemical must be carefully controlled because predatory mite populations can be decimated by applications of chemical pesticides through the growing season, potentially resulting in greater damage levels by the Grapevine rust mite.

Grape Blossom Midge

The grape blossom midge lays eggs in buds of grapes, and the maggots then damage the blossoms, preventing fruit development. These insects spend very little time actually on the plant. Following budding, the insects drop to the ground to pupate, and lie dormant in the soil until the following spring. Chemical control can harm populations of predator mites

Grapevine Flea Beetles

Grapevine Flea Beetles (*Altica chalybea*) are small 3-4mm beetles, blue-black in colour. The beetles jump.

The adults feed on the unfolding leaf buds and the grape buds; larvae feed on the upper surface of the leaves, skeletonising them and also attack the flower clusters.

Biological insecticides may afford some control.

Birds

Birds can be a pest, taking fruit from vines. However, some bird species can also play an important part in insect control. The viticulturist should aim to encourage insectivorous birds that do not damage fruit, whilst discouraging birds that damage fruit.

Birds can cause damage to grapes which in turn leads to oxidation and development of disease. On large vineyards (over 20ha or 50 acres), there is less likelihood of damage, because there are fewer nesting places in the vicinity (birds are territorial, and many problem birds do not travel far from their nests).

Control methods include techniques to deter birds such as scarecrows or visible deterrents in the vines, scare-guns etc., as well as bird netting. Whilst it is possible to poison birds, this is usually illegal and can also be dangerous. One method used on some vineyards is to plant an early ripening, long-yielding 'catch crop' vine. Birds feed in the catch crop undisturbed, and thus it is more attractive to them than the commercial crops. Ideally catch crops should be located away from the main vineyard.

Large Animals

Large animals usually pose most threat to young plants. Vineyards should be fenced off from stock and wildlife, although fencing is not always effective. Electric fencing can be useful in some situations. Sturdy plant guards can also provide protection from large animals.

Disease Control in Vineyards

Grapes are susceptible to many diseases including fungal diseases, bacterial diseases and a number of viruses. Various control methods are used, including propagation of resistant varieties, use of hardy hybrids and rootstocks, pruning and training techniques, and various chemicals. In extreme cases, soil may be fumigated.

Fungal Diseases

There is a wide variety of fungal diseases that affect grape vines. The effects of these diseases range from rot in the roots, wood and fruit, to spots and discolouration in leaves and fruit. Extreme infections may result in death. Many fungi thrive in mild, moist weather, hence site selection is important. A vineyard that has good air circulation and is situated in a dry, warm climate will experience fewer fungal problems than a vineyard in a humid or damp climate. Control usually relies on controlling disease vectors, effective sanitation, pruning and training techniques that improve air-flow, and fungicidal sprays.

Sooty Mould

Sooty mould appears as a black coating on the surface of plant tissue made up of dark coloured threads of fungi. These fungi are not parasitic but grow on the "honey dew", the sugary deposit exuded by insects such as mealy bugs, aphids or scale: the fungi do not directly affect the plant, but as a dense layer of mycelium forms on the leaves, the plant's ability to photosynthesise is inhibited. Sooty mould does indicate the presence of pest insects. Sooty mould is controlled by spraying to control the problem insects, thus stopping the production of honeydew.

Summary

- Sooty mould indicates the presence of pest insects.
- Sooty mould stops photosynthesis occurring in plant tissue, by stopping light reaching the green tissue underneath it.

Management

- Spray to control the problem insects, thus stop the production of honeydew.
- Some fungicides may kill the mould, but if the honeydew remains, the fungus will regrow.

Botrytis Bunch Rot

Botrytis spp. are grey moulds. These grey moulds are serious, widespread diseases worldwide on flowers, vegetables, ornamentals, fruits and some field crops. They can cause serious losses in susceptible grape varieties. Although all grape varieties are prone to this fungus, tight-clustered varieties are most prone to serious damage. Fruit that is damaged by birds etc. is frequently also more susceptible to infection.

Botrytis is caused by the fungus *Botrytis cinerea*. In humid conditions a noticeable fluffy, cobwebby, whitish-grey or light brown growth is produced that is characteristic of Botrytis disease. Good air circulation is an effective preventative control mechanism. Damage to berries should be prevented. In commercial situations growth regulators designed to make the berry bunches less tightly clustered are used, and fungicide sprays applied at appropriate times can provide significant control.

Cultural Control

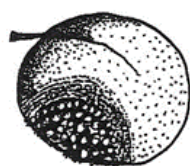
- In greenhouse grape production humidity should be reduced by improved ventilation and increased heating.
- Berry damage should be prevented.
- Select resistant grape cultivars.
- Promptly remove infested debris.

Chemical Control

Fungicide sprays applied at appropriate times will control botrytis.

Rots

Rots are diseases that cause decay or disintegration of plant tissue. Rots in grape vines include collar rot, black rot, root rot, and a variety of others. Grape vine rots are usually caused by fungal infections. Rots can affect roots, wood, leaves, or fruit. Rots can be spread by wind, irrigation and rain, and in some regions can pose a significant risk to commercial production. Black rot, for example, is one of the most serious problems of cultivated grapes in the eastern United States of America, particularly in humid areas. Good sanitation, involving removal of all infected plant parts, along with strategic annual fungicide sprays is necessary for control.



Rots

Black Rot

Black rot is one of the most serious diseases of grapes in the eastern United States. It is caused by the fungus *Guignardia bidwellii* and can affect all green parts of a vine. The most serious effect is on the fruit and severe infestations can result in total crop loss.

Reddish-brown spot appear on the upper leaf surface and the centre of the spot turns tan to light brown. Small black pimples form in a loose ring at the centre of the spot.

The grapes become black, wrinkled and mummified.

The fungus over-winters in mummified grapes on the soil or on old berry clusters left on the vine. Spores infect the leaves, blossoms and young fruit using the water from spring rain, or irrigation as a conduit. Infections depend on temperature and how long foliage is wet.

Cultural Controls

Infected prunings and mummified berries should be removed. Resistant cultivars should be chosen.

Chemical Controls

If black rot was a problem the previous growing season, use early preventive fungicidal sprays. If too many infections occur, chemical controls will be unsuccessful against fruit rot later in the season.



Mildew

Mildew are caused by fungi and will occur during warm wet conditions at any time during the growing season. Mildew causes young leaves and tip growth to become distorted and appear grey or discoloured.

Several different types of mildew occur on grapes, including downy mildew, powdery mildew and black mildew.

Downy Mildew

Downy mildew is caused by the fungus *Plasmopara viticola*. It is an obligate parasite - that is, it must parasitize a host plant to survive. High humidity, damp conditions and a film of water on plant tissues increases susceptibility of host and development of mildew. The upper leaf surface develops yellow spots, while a grey mould begins to develop on the underside, directly below the spots: later the lesions turn brown and the under surface turns dark grey: distortion of the leaves and stems may occur.

Cultural Controls

Keep soils well drained and clear vine debris, such as fallen leaves to remove over-wintering spores. Irrigate from underneath in hot, humid or moist conditions.

Chemical Controls

Fungicides are the most important control measures.

Powdery Mildew

Powdery mildew is probably the most common, conspicuous and widespread of diseases: it is easily recognisable. Infections are characterised by spots or patches of white to greyish, powdery, cobwebby growth on young tissues or entire plants. Causes young leaves and tip growth to become distorted and appear grey.

Cultural Controls

Keep soils well drained and clear vine debris, such as fallen leaves to remove over-wintering spores. Irrigate from underneath in hot, humid or moist conditions.

Chemical Controls

Fungicides are the most important control measures.

Black Mildew

Appearance is similar to powdery mildew, only a dark colour: cobweb-like to powdery black patches appear on the upper surfaces of mature leaves. As the infection develops, the upper leaf surface becomes densely coated with a black powdery growth of fungal hyphae. Sometimes confused with "sooty mould", black mildew does not develop on the sticky exudates known as honey dew, which are secreted by insects, as does the sooty mould. Sooty mould also differs from black mildew because it is saprophytic (feeds by absorbing dead organic matter) and superficial. Several fungi cause black mildew.

Management of mildew problems is usually achieved by good plant culture, such as pruning to allow air flow, irrigating in a way that minimises humidity in the crowns of plants, and by good vineyard hygiene. No chemical controls are recommended for black mildew.

Eutypa Dieback

One of the most destructive diseases of the woody tissue of grapes was once known as Dead Arm Disease. It is now established that this is in fact two separate diseases that often occur simultaneously. Eutypa dieback (caused by the fungus *Eutypa armeniacea*) is the canker and shoot-dieback phase of what was formerly called Dead Arm Disease.

In the early stages, cankers form around pruning wounds. Concealed by old, dead bark, the cankers are often hard to detect. As new shoots develop above the cankered area, the internodal length is reduced and the growth is stunted. Leaves are reduced in size, yellow and crinkled. The foliar symptoms worsen each year, until eventually no shoots form on the diseased "arm". Bunches



on affected shoots are a mixture of large and small berries

The disease is caused by the fungus *Eutypa lata* and is unlikely to occur where rainfall is less than 250mm (10in) per year. Most common occurrences are in areas where rainfall or equivalent overhead irrigation precipitation exceeds 500-600mm (20-24in) per year. This also applies to vineyards which receive the equivalent in overhead irrigation.

The fungus reproduces in areas where 350-500mm (14-20in) or more of precipitation occurs annually.

There are no resistant grapevine cultivars are susceptible to *Eutypa* dieback. No chemicals routinely used in vineyards to control fungus or fungus disease on vines are effective against *Eutypa*. Grenache and Shiraz grapes are particularly susceptible to *Eutypa* dieback, but it is commonly seen in many different types of grape.

Cultural Controls

Some prevention opportunities arise from timing and application of pruning techniques. Pruning wounds can be vulnerable for 2-4 weeks, so estimating the weather patterns for a month after pruning can give an indication of the likely chance of *Eutypa* dieback infection. Tools must be cleaned between cuts. Where infection is observed, cut off the diseased arm, cutting far enough back until no discoloured wood is seen; then cut back an additional 5cm (2in). The wound must immediately be soaked in fungicide. Most importantly remember, that since there is no cure for the disease, avoid wounding vines for any reason in wet weather and particularly cold, wet weather.

Chemical controls

Infections occur on freshly made wounds. Wound protectants should be used on any cut with a diameter of more than 1.5cm (½in) and should be applied immediately after pruning.

Bacterial Diseases

Some countries, such as New Zealand and the United Kingdom, have lower incidences of bacterial diseases on grape stock. Other countries, such as United States, face significant problems with these bacteria, and resistant vine-stock is necessary. The warm, humid climate of the southern regions of the US is particularly favourable to the spread of infectious bacterial diseases in vines, particularly Pierce's Disease. The disease is less prevalent where winter temperatures are colder, such as at higher altitudes, northern latitudes and inland from ocean influences.

Pierce's Disease

Pierce's disease is a bacterial infection that affects the xylem tissue of grapevines, leading to vine death. It is the most significant obstacle to the growing of European-type grapes in the United States. The disease, caused by the bacterium *Xyella fastidiosa*, is spread in infected propagation material and by xylem-feeding insects. The Glassy-Winged Sharpshooter (*Homalodisca coagulata*) is a vector of the bacterium, as are the blue-green sharpshooter (*Graphocephala atropunctata*), the green sharpshooter (*Draeculacephala minerva*) and the redheaded sharpshooter (*Carneocephala fulgida*). Other sucking insects, such as grape leafhoppers, are not vectors.

Symptoms of the disease include death of the leaf margins, leaf drop, irregular maturation of new canes, loss of vigour, stunting, delayed bud-burst and die-back.

There are no consistently effective cures for infected plants. Insecticide treatments aimed at controlling the sharpshooter and other vectors help to reduce the incidence of Pierce's disease. Some grape varieties such as Chardonnay and Pinot Noir are particularly susceptible. The main recommendations for control are the early detection of the disease followed by vine removal in vineyards where the glassy-winged sharpshooter is observed.

Elsewhere, removal of vines that have had Pierce's symptoms for more than one year, or have been infected long-term is a key management tool. Although changes to the pruning regime, e.g. cutting back to a few centimetres above the graft union of vines where foliar symptoms are mild, may appear to eliminate the disease, symptoms will reappear in the second year.

The glassy-winged sharpshooter feeds much lower on the cane than other sharpshooters and in some areas, such as California, infections introduced by this vector may survive the winter to cause chronic Pierce's disease. Insufficient is yet known about the relationship between the glassy-winged sharpshooter and Pierce's disease.

Viruses

Viruses are small microscopic organisms which live inside the bodies of other organisms. They are parasites and can have a wide variety of different effects on the organism they infect. One of the most common symptoms in plants is a change of colour in leaves/and or flowers. Infected leaves frequently show light green or yellow patches (i.e. a variegated effect) due to interruption of chlorophyll production with a consequent reduction in photosynthesis. This type of infection can cause reduction of crop yield or quality, or a general stunting of the plant.

One of the most severe effects of virus would be death, although this is not common (virus can only live in a host organism while the host is alive; if the host dies, the virus dies and hence eliminates itself). Growth patterns can be disturbed and changed by virus. In some cases stunting (mild or severe) will be the only obvious effect. In other cases, virus can cause distortion in the growth (e.g. twisting, blistering or other distorted formations in leaves, stems, roots or flowers). In extreme cases, leaves can be reduced to a central midrib (i.e. no leaf blade at all).

Flowering and seed production can be stopped completely by a virus. Virus can also induce leaf rolling, leaf yellowing, plant wilting, and changes to the physiological processes in the plant so that some functions of a cell's metabolism cease completely. One or several of these symptoms might occur.

Viruses are not as easy to control as most other diseases. Once a plant cell is infected with a virus, the only way to eradicate the virus is by killing that plant cell. In the case of virus, generally speaking, prevention is the only cure. Control of insects and other organisms such as aphids, fungi and nematodes can limit virus spread, since these organisms are capable of spreading viruses. Use of planting stock that is 'certified free of known viruses' can minimise problems from the outset. If a plant is infected with virus remove and burn the plant.

Viruses that affect grapevines include fan leaf virus, leaf roll virus and shoot necrosis virus.

Leaf Fan and Leaf Roll Viruses

These two viruses infect and are spread between vines and rootstocks during propagation. Frequent virus testing, scrupulous hygiene and good nursery practice are the only safeguards against infection.

Downward turning leaf edges are the characteristic symptom of leaf-roll virus; in red grape varieties, the leaves also turn brilliant red. The effect of the virus is to slow down the grape's accumulation of sugar which means that by harvest time the acid and sugar balance is spoiled and quality is diminished. The virus does not kill the vine but yields will be reduced.

There is no treatment other than to remove and destroy infected vines. Prevention can be achieved using certified virus-free stock.

Organic Culture of Grapes

Organic culture means growing produce without the use of artificial chemicals. 'Organically-grown' produce is very popular in many countries, and grape products are no exception. Wines, table grapes, juice and dried fruit are grown organically. Quality produce that complies with recognised organic certification schemes will command premium prices.

It is important to realise that crop losses are likely to be higher, especially in the short term, compared to grapes sprayed with chemicals. Blemishes will also be more common, although minor blemishes may be acceptable to some sectors of the market.

Organic Pest and Disease Control

Without chemicals, you will always have some damaged fruit, but if you can accept that, the following controls are safer than chemicals and will help minimise problems.

- Plant chives or garlic in a circle around the butt, under your grape vines. This is a long-term treatment, with strong evidence to support its effectiveness, particularly under peaches where it controls peach leaf curl. Effectiveness with grapes is not yet

fully known.

- Remove and burn dead or diseased fruits or leaves from the vine or the ground as soon as you see them.
- Spray or throw soapy water from the washing machine (i.e. grey water) over the vines frequently (this helps to suffocate insects).
- Use traps to catch fruit fly, aphids, and other pests (including possums and rodents).
- Grow resistant varieties. Grafting is often used.
- Promote predators of pests. Release predators which attack pests; for example, predatory mites can be purchased which attack pest mites. Attract predators to the vines; for example, flowering plants growing near the vines will attract wasps and other predators which attack and kill certain pests.
- Allow poultry to roam ('free range') beneath your vines to help control pests, and to add valuable manures to the soil.
- Monitor levels of magnesium and phosphorus in the soil – vines deficient in these nutrients are more susceptible to Phylloxera. Rock phosphate is an organic alternative to superphosphate as a source of soil phosphorus.
- Nicotine tea sprayed onto leaves helps to destroy aphids and other leaf-chewing insect.
- Plant French marigold (*Tagetes* spp.) and nasturtium (*Tropaeolum majus*), beneath vines to help repel nematodes and insect pests respectively.
- Monitor fungal diseases and insect infestations by growing roses at the end of each row of vines. Roses are susceptible to many of the same problems as grapes – a quick routine check of the roses will alert you to problems that may be occurring (or about to occur) in the vines. Early detection reduces the need for resorting to chemical controls. This method has been used for centuries in French vineyards (long before synthetic sprays were invented!).

Birds

Birds can be a nuisance in vineyards, eating or damaging the grapes, but there are simple and effective ways of dealing with this problem:

- Netting
- Scarecrows
- Plastic bags hanging in the trellis
- Cats in a garden or small vineyard will help scare the birds away (but give the birds a fair chance – plenty of open space around the vines will make it difficult for the bird to be caught, and difficult for the bird to get near the vines without being seen).

Remember though, that birds also eat many pests of your fruit trees and vines.



Other Non-Chemical Controls

As described in the previous section, pests that attack grapes include aphids, leaf hoppers, mealy bug, nematodes, mites and birds. Diseases may include downy mildew, botrytis and powdery mildew.

The root aphid *Phylloxera* has caused major destruction of non-resistant grapes varieties in many parts of the world. In these areas replanting is often undertaken by budding/grafting onto *Phylloxera*-resistant root stocks. In many areas strict quarantine regulations apply to the transfer of grape plant material to prevent the spread of grape pests and diseases, particularly *Phylloxera*. These regulations should be strictly complied with.

- Monitor levels of magnesium and phosphorus in the soil – vines deficient in these nutrients are more susceptible to *Phylloxera*. Rock phosphate is an organic alternative to superphosphate as a source of soil phosphorus.
- Cover plants with netting as fruit nears maturity to deter birds.
- Fungal diseases are less likely if there is a good flow of air around plants, so keep plantings in the open and well-spaced. Sulphur sprays and dusts also offer some control of fungal problems.
- In areas with heavy soils, collar rots sometimes occur near the base of the trunk, particularly if the trunk has been damaged (e.g. by machinery). In this case the damaged area should have any soil or debris removed away from the trunk to expose the area to the wind and sun. Lime sulphur can also be applied to give some degree of control.
-

Companion Plants for Grapes

Companion planting is based on the idea that certain plants promote or inhibit the growth of other nearby plants. Some growers have found that grapes can benefit from nearby plantings of blackberry, raspberry, hyssop, legumes (e.g. peas and beans), chives or garlic. It has also been suggested that grapes should not be grown near cabbage, radish and other brassicas.

Environmental Problems

Extreme heat, cold, wind, rain, frost, shade, pollution and other environmental problems can have disastrous effects on vines.

Common Environmental Problems

Soil and Air Pollution

Soil can be contaminated from chemicals, such as fertilisers, pesticides, and residual herbicides. Vines growing in contaminated soil will fail to thrive and will have discoloured and burnt foliage. Leaching the soil with water may help to remove the contaminant but often the only long-term solution is to dig out the contaminated soil and replace it with fresh soil.

Plants that are suffering from over-fertilising will also develop burn marks on the foliage and root tips. Fertiliser burn is more likely in hot weather when many fertilisers become more soluble and the chemicals are released quickly into the soil.

The most common symptoms of air pollution are burning and discolouration of leaves and growth tips. The effects will be worse in badly polluted areas, so if you suspect pollution from a nearby factory, look at plants growing closer to the factory - they should show more dramatic signs of pollution.

Drainage Problems

Too much water around roots starves young roots of oxygen and encourages the growth of fungal root rots. Plants growing in damp soil will develop black or brown root tips and yellow leaves. The leaves will eventually drop, and the plant may die.

Treatment:

- Lay drainage pipes in damp areas, preferably before the vines are planted.
- Aerate the soil with an aeration fork.
- Apply gypsum to break up the clay particles in the soil.

Frost Damage

Frost damage generally appears as dead areas of foliage and stems. The affected parts turn black, and may go watery. Frost can affect leaves, growth tips, fruit, flowers and buds.

Symptoms begin to show within hours of the frost occurring. Areas on the outside of the plant are most affected (leaves closer to a wall or other plants are less affected). Frost damage to fruit buds or flowers can cause a significant reduction in fruit setting.

Treatment:

- Don't remove burnt foliage until danger of further frosts has passed. (Dead damaged tissue should be cut off to prevent infection spreading as soon as frosts finish. Premature pruning will stimulate new shoots which will be particularly tender and exposed to frost damage.)
- Cover with hessian or some other physical protection.
- Create air movement to stop frost settling (e.g. in a frost-prone greenhouse, a fan might be left on overnight when there is a frost warning).
- Use a heat source (e.g. a kerosene fire or fan heater) on frost warning nights. Even in the open, a small fire creates air movement which is often enough to prevent frost settling.
- Put sprinklers on for a few hours just before dawn when the chance of frost is greatest (i.e. the slightly higher temperature of the water is often enough to prevent frost damage). This is a specialised technique and should only be used following expert advice.



Frost Pot is lit to raise vineyard temperature on cold nights.

Hail

Hail can severely damage fruit, flowers and foliage. The damage is obvious, causing denting and tearing of plant tissue. Damaged tissue cannot be repaired and is normally cut off. Hail netting may be an option for protecting small vineyards in areas that are prone to severe summer hail storms.

Shade

Areas of a vineyard may become increasingly shaded as nearby trees grow (e.g. windbreak trees), without you noticing the change. Growth can become elongated and less bushy as vines stretch toward the light. Plant vigour, flowering and fruit set will soon deteriorate. The soil may remain moist in over-shaded areas, and algae or moss may grow on the soil surface.

Treatment:

- Remove some of the large trees.
- Thin out branches in the canopies of large trees to let more light penetrate.
- Replace evergreen trees with deciduous trees to let more light in over winter.

Temperature

Cold temperatures slow down growth and reduce flowering and fruit development. Extreme cold will kill parts or all of a plant very quickly.

In high temperatures, the plant will wilt and leaves will become scorched. Young leaves and growth tips will quickly wither and blacken. In extreme heat, the plant will die rapidly (within a few hours of exposure to high temperatures).

Treatment:

- Mulching reduces temperature extremes in the soil.
- Watering and shading can reduce temperature extremes.

Wind

Strong wind causes foliage to dry out faster than water can be taken up from the roots. This causes wilting, particularly with more susceptible newly-planted vines. In established plants, wind can cause drying out and death of the more tender and exposed parts of a plant. Damage will be worse on growth tips and tender foliage.

Treatment:

- Erect or plant a windbreak.

Water Management

Water resource management is becoming increasingly important for agricultural producers. Many vine-growing regions are located in areas with dry climates, low fertility soils and scarce water supplies. Recurring droughts, dwindling fresh water supplies and a general decline in the quality of available water means that viticulturists in low rainfall areas increasingly need to understand how to manage water efficiently to ensure they remain viable. Growers in climates with reliable rainfall will also benefit by conserving and managing the water on their property. Quality water supplies will become scarcer as populations increase and urban developments encroach on agricultural land. Pollution from urban runoff, contamination resulting from increased sedimentation and fertiliser runoff, and changes in rainfall patterns caused by global warming, are just some of the problems that growers will need to deal with in the future.

Runoff

Runoff is the term applied to the movement of water (especially rain) when flooding occurs. In the process of moving towards the line of least resistance, water builds up speed and begins to erode the surface soil. Runoff may affect both vineyards on slopes and those lying at the bottom of steep hills. During heavy rain, slopes may be scoured and valuable topsoil washed downhill. Low-lying vineyards may experience flooding and dumping of sediments and other debris.

Control

Runoff can be controlled with a number of strategies:

Cultivating on the contour

This refers to cultivating with the contour of the land rather than across the slope. By doing this, water is slowed down by the ploughed ridges, giving it more time to soak into the ground.

Building contour banks

Contour banks should also be designed to follow the land contours. Their purpose is to direct the flow of water into grassed waterways. The grass reduces the speed at which water flows over the ground, and reduces erosion and sediment dumping down slope. Runoff flowing down the grassed waterways is channelled into catchment areas such as streams and dams without the loss of soil. This can be an effective way of catching runoff and directing it to a suitable storage site (dam) for later use.

Strip cropping

The purpose of strip cropping is to disperse and slow the movement of flowing water. This tends to restrict erosion damage as the water cannot build up in volume and hence speed as it is dispersing. The slower movement of water allows more time for it to infiltrate into the soil, where it can be utilised by the plants growing there.

Water-saving Measures

The watering requirements of your grapes can be minimised in the following ways:

- By choosing grape varieties that best suit the local climate.
- By maintaining a well-structured, fertile soil (appropriate to the plants selected).
- By irrigating in the cool of the day.
- By using an irrigation system that minimises wastage, for example, micro-irrigation systems (e.g. trickle systems or soaker pipes) on long-term crops such as fruit trees and grapes.
- By slow, thorough watering. A thorough deep watering once or twice a week will be more effective than light watering every day or two.
- By avoiding irrigating, where possible, on windy days.
- By reducing excess evaporation. This can be achieved by keeping bare soil covered where possible. Mulches, as well as reducing weed growth will reduce evaporation, as do windbreaks.

Grape Clones and Varieties

Selecting proven clones and varieties is an important strategy for ensuring good vine performance and resistance to pests and diseases.

The term 'clone' refers to the original parent plant from which a group of plants was propagated. For example, the plants from a particular variety of Sultana growing in one vineyard may all be propagated from a plant bought from one nursery; while the plants in another vineyard may have come from a different source. These two groups of Sultana vines, while still Sultanas, have originated from different sources and, as such, are considered to be different clones.

The differences between clones lie in their genetic makeup. In many cases, different clones of the same variety may be very similar in their genetic makeup and perform just as well as each other in the vineyard. In other cases, those small differences may make a difference – one clone might be slightly better for your situation than another. For example, one clone of Chardonnay may resist certain pests or diseases a little better than another clone of Chardonnay. If you find that a particular clone is performing exceptionally well in your situation, it may be wise to only use that clone in future.

Cloning means that the source of grape vine propagation material is very important. For reasons such as those explained above, you should try to be aware of the clonal origins of plants you grow and, if you want the best results from your vines, use not just proven varieties, but also use proven clones.

In Australia, the USA and Germany, there are many research programs in clonal selection which are predominantly based upon strains/varieties that are virus-free, although other selection factors include mutations which may prove beneficial. (Mutations occur when the genetic units of the cells dramatically change their function and affect the plant adversely. They may arise at random.

They are also known as bud sports.)

There are various industry bodies around the world that may be able to supply cuttings or plants from new cultivars and selected clones, such as:

- Research stations
- Government agriculture departments
- Industry associations

In addition, clones are often available as rooted and grafted vines through nurseries accredited by industry associations.

